THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **Shaping the Earth's Surface, Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30–40 km	thicker than oceanic crust
2	**(a)** core	solid, hot, spinning
3	**(c)** divergent	new crust forms
4	**(a)** earthquakes, volcanoes and mountains	tectonic activity
5	**(a)** deposition	builds landforms
6	**(d)** 21%	second most abundant
7	**(a)** free of clouds and weather disturbances	stable layer
8	**(d)** decreases with altitude	coldest layer
9	**(b)** increasing altitude	mountains are cooler
10	**(d)** high pressure to low pressure	wind

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Breaking down of rocks on or near the Earth's surface into smaller pieces by physical, chemical or biological action (2).	2
12	Erosion removes and transports material; deposition lays it down (1+1).	1+1
13	Water falling from the atmosphere (1); rain and snow (1).	1+1
14	Weight of air above a place (1); wind flows from high to low pressure (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Debris carried and deposited by glaciers (1.5); lateral and terminal moraines (1.5).	1½+1½
16	March to May with rising temperatures (1); hot dry winds like loo in the north (1); pre-monsoon showers in the south (1).	1+1+1
17	Tourism and trekking, harbours in fjords, fresh water supply (3).	3
	OR: Barchan crescent-shaped; longitudinal; star dunes (1+1+1).	1+1+1
18	June to September (1); moist winds bring heavy rain to most of India (1); essential for agriculture (1).	1+1+1
	OR: October to December (1); retreat of monsoon, clear skies in the northwest (1); cyclones occur on the east coast (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Cirque, arete, U-valley, fjord, moraine (2). (ii) Cirque and arete erode; moraines deposit (2). (iii) Tourism, fresh water, harbours (1).	2+2+1
	OR: (i) Fjord (1). (ii) The sea flooded a glacial U-shaped valley (1). (iii) Cirque or moraine (1). (iv) Norway (2).	1+1+1+2
20	(i) They noted lunar mansions and signs to predict rainfall (2). (ii) Krishiparashara and Brihatsamhita (1). (iii) It helps farmers and complements science (2).	2+1+2
	OR: (i) To improve forecasting and climate readiness (1). (ii) IMD (1). (iii) Plan sowing and harvest and save crops (1). (iv) Conserve water and energy (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **Shaping the Earth's Surface, Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** delta	e.g. Sundarbans
2	**(b)** aretes	glacial erosion
3	**(b)** barchan dunes	arid regions
4	**(c)** contour trenching	soil conservation
5	**(d)** generation of electricity	hydropower
6	**(c)** Arab traders	mausim = season
7	**(a)** nakshatras (lunar mansions)	Brihatsamhita
8	**(a)** western disturbances	winter rain
9	**(c)** floods	excess rainfall
10	**(b)** temperature	°C

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Running water, wind, glaciers, sea waves (2).	2
12	A looped bend in a river (1); formed as the slow river erodes the outer bank and deposits on the inner bank (1).	1+1
13	Sea breeze blows from sea to land in the day; land breeze blows from land to sea at night (1+1).	1+1
14	Seasonal reversal of wind direction (1); the Arabs (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wind deposits fine dust (1.5); it forms fertile soil good for farming (1.5).	1½+1½
16	Their temperate climate gives autumn and spring (3).	3
17	Landscape formed by dissolving of limestone by water (1); caves, sinkholes, stalactites, stalagmites (2).	1+2
	OR: Contour trenches slow runoff and increase infiltration (1.5); terraces break the slope so water flows slowly (1.5).	1½+1½
18	Supports agriculture (1); fills rivers and reservoirs (1); affects food prices and incomes (1).	1+1+1
	OR: A weak monsoon causes droughts (1.5); a heavy one causes floods (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Dunes, mushroom rocks, loess plains (2). (ii) Barchan, longitudinal, star (2). (iii) Fertile deposit of wind-blown dust (1).	2+2+1
	OR: (i) Limestone landscape from dissolving by water (1). (ii) Dripping water leaves calcium carbonate on roof and floor (2). (iii) Caves such as Borra or Belum caves (2).	1+2+2
20	(i) Greenhouse gases from fuels and deforestation (1). (ii) Floods, droughts, glaciers melting (2). (iii) Renewable energy, afforestation, international agreements (2).	1+2+2
	OR: (i) Humidity: water vapour; precipitation: water falling (2). (ii) Rain, snow, hail (1). (iii) Moist winds rise, cool and condense (2).	2+1+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **Shaping the Earth's Surface, Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5–7 km	thin and dense
2	**(d)** huge slabs of solid rock that move slowly	carry continents and oceans
3	**(c)** transform	earthquakes
4	**(b)** landform	shaped by processes
5	**(a)** V-shaped valleys and waterfalls	erosion dominates
6	**(b)** near the Earth's surface	decreases with altitude
7	**(c)** mesosphere	extends to 80 km
8	**(b)** weather	short-term
9	**(c)** they receive direct sun rays	latitude
10	**(c)** sea breeze	land is warmer

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A landform at a river mouth formed by deposition of sediments (2).	2
12	Stalactites hang from cave roofs; stalagmites rise from the floor (1+1).	1+1
13	Winter, summer, monsoon and post-monsoon (2).	2
14	The withdrawal of monsoon winds from India during October to December (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Integrated farming of Nagaland using earthen bunds and ponds on hill slopes (1.5); conserves soil and water (1.5).	1½+1½
16	Weather monitoring and forecasting (1); cyclone and flood warnings (2).	1+2
17	Sindhu-Saraswati civilisation used terracing, bunding, dams and canals (1.5); Arthashastra gave guidelines on land assessment (1.5).	1½+1½
	OR: Loss of roots exposes soil (1); runoff and wind carry topsoil away (1); floods and landslides increase (1).	1+1+1
18	Save energy, use water carefully, plant trees, use public transport (1+1+1).	1+1+1
	OR: Rain, snow, hail, sleet (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Soil erosion (1). (ii) Roots no longer bind the soil (1). (iii) Terracing, contouring, afforestation (2). (iv) Lower yields and floods (1).	1+1+2+1
	OR: (i) Loss of topsoil, unsafe land, damaged beaches (3). (ii) Terracing, contour trenching, bunding, planting trees (2).	3+2
20	(i) The inland city (1). (ii) Land heats and cools faster than water (1). (iii) Higher places are cooler (1). (iv) Clouds reduce day heating and night cooling (2).	1+1+1+2
	OR: (i) Thermometer, barometer, rain gauge, wind vane, hygrometer (2). (ii) Temperature, pressure, rainfall, wind direction, humidity (2). (iii) Climate is the long-term average (1).	2+2+1

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ANSWER KEY & MARKING SCHEME — SET D

Chapters 3 & 4: **Shaping the Earth's Surface, Atmosphere and Climate** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** fertile alluvium	deposited by floods
2	**(b)** moraine (till)	glacial deposits
3	**(a)** stalactites	karst topography
4	**(b)** terracing	hill farming
5	**(b)** wind-blown fine dust	wind deposition
6	**(a)** unequal heating of land and sea	low pressure over land
7	**(d)** winter, summer, monsoon and post-monsoon	four seasons
8	**(c)** Weather Ready and Climate Smart	global leader
9	**(b)** autumn and spring	temperate
10	**(b)** differences in air pressure	winds

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A deep narrow inlet formed when the sea floods a glacial valley (2).	2
12	Terracing, contour trenching, bunding (1+1).	1+1
13	A mission to make India Weather Ready and Climate Smart with better forecasting (2).	2
14	Krishiparashara and Brihatsamhita (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fertile alluvial soils support agriculture (1.5); water, fishing and transport are available (1.5).	1½+1½
16	High humidity makes sweat evaporate slowly so we feel hot (1.5); low humidity makes the air dry (1.5).	1½+1½
17	Many caves, rivers and mountains have religious or cultural significance; e.g. Himalaya and Ganga (3).	3
	OR: Explains earthquakes, volcanoes, formation of mountains and ocean basins (1+1+1).	1+1+1
18	A strong monsoon with a rainy and a dry season (1.5); located in the monsoon region (1.5).	1½+1½
	OR: Long-term change in climate patterns (1); more floods, droughts and melting glaciers (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Internal build mountains and volcanoes; external wear down and level them (3). (ii) Plate movement, volcanoes; rivers, glaciers (2).	3+2
	OR: (i) Helps explain hazards and resources (1). (ii) Floodplains for farming, waterfalls for power, glacial valleys for tourism (3). (iii) By sustainable use and conservation (1).	1+3+1
20	(i) It traps part of the Sun's heat (2). (ii) Gases like CO_2 and methane (1). (iii) Less fossil fuel use, more trees (2).	2+1+2
	OR: (i) Different latitudes, land and water absorb insolation differently (2). (ii) Warm air rises giving low pressure and cooler air sinks giving high pressure; air flows between them (2). (iii) Sea breeze or land breeze (1).	2+2+1

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